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SECTOR AT A GLANCE 07

Rural: Tap Water Supply in India

WASH in India

DRAFT

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RURAL: TAP WATER SUPPLY IN INDIA

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

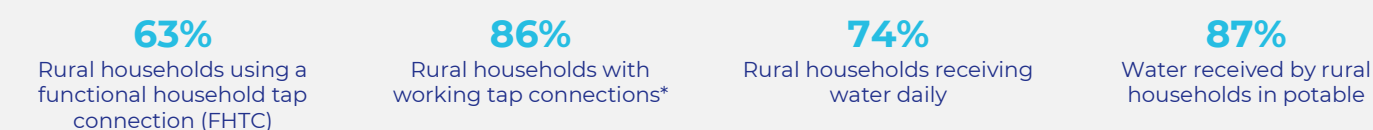
The Jal Jeevan Mission aims to ensure an adequate quantity of safe drinking water for every rural household in India by 2024. Under this mission, significant progress has been made toward providing each person in rural India with 55 litres of water per day, especially by improving the availability of tap water connections.¹ Nearly 86% of rural households were assessed as having tap connections, and 63% had a functional tap connection based on parameters related to regularity, water quantity, and water quality. Thus, this has the potential to drive significant improvements in health, education, economy, and gender outcomes.²

However, there are still challenges that need to be addressed: uneven coverage in several states, irregularity in many locations, water contamination in pockets, poor understanding of water potability by citizens, and insufficient coverage of water committees at the village level. It will be important to address these moving forward.

This note captures key trends related to rural water supply covering both water access and water supply, opportunity areas, challenges, and also the institutional architecture of Ministries, Departments, and Policies that are key to rural water supply.

It ends with a few ideas that could catalyse progress going forward.

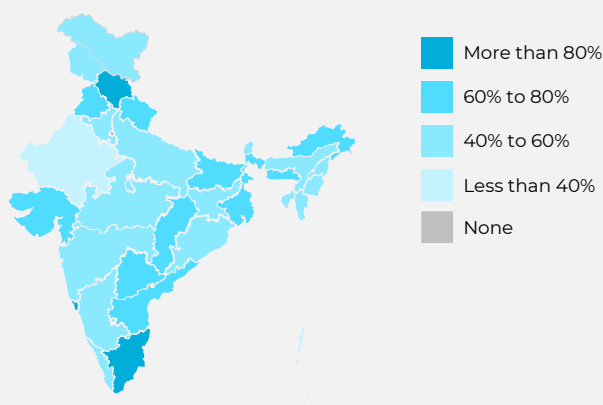
Figure 1: Key rural tap water supply statistics



Source: Department of Drinking Water and Sanitation, Functionality Assessment of Household Tap Connections, 2022

Note: * Working tap connections involve those households which received water through FHTC at least once in the last 7 days

Figure 2: Percentage of households with functional tap connections by state and UT (2022)



Source: Ministry of Jal Shakti, Functionality Assessment of Household Tap Connections, 2022

Note: "Functionality" has been computed as the intersection of Adequate Quantity, Fully Regular Supply and Potable (Quality) for households wherein water supply was available at the time of the survey, i.e., 2,59,151 HHs.

¹ Jal Jeevan Mission, Functional Household Tap Connection (FHTC) in every rural home, accessed 17th June 2023

² NITI Aayog, CWMI 2.0, 2019; Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022; World Bank, Rural Population - India, accessed July 2023

Key trends

This section dives deeper into India's rural tap water supply landscape, outlining statistics and trends on:

1. Water Access
2. Water Quality

Water Access

Water access, which involves the establishment of functional tap water connections in households and institutions such as schools, AWCs (Anganwadi Centers), GPs (Gram Panchayats), and CHCs (Community Health Centers), has improved significantly under the Jal Jeevan Mission.

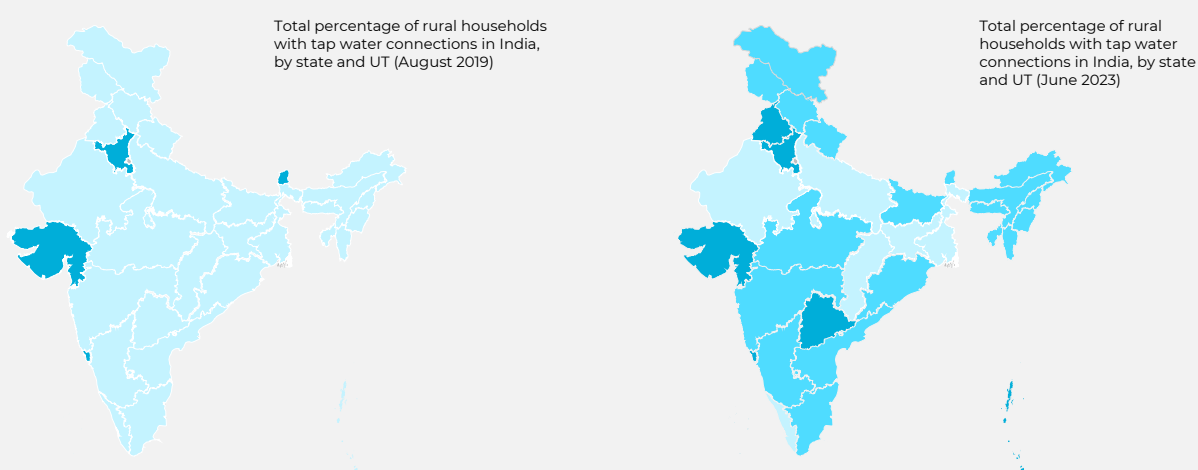
Tap water connections have increased significantly under JJM:

Under JJM, tap water connection coverage improved from ~17% of rural households (32 million) in 2019 to ~63% of households (123 million) in June 2023.³ In 2020, a 100-day campaign was launched to provide tap water supply in schools and AWCs.⁴ In less than ten months tap connections had been provided in 66% of schools, 60% of AWCs, and 69% of GPs and CHCs.⁵ As of June 2023, coverage stood at ~89% schools, ~85% AWCs, and ~77% GPs and CHCs.⁶

A majority of these tap water connections are fully functional and have regular water supply:

A Jal Jeevan assessment report from 2022 notes that out of 86% of sampled households that reported having working tap connections, 62% stated that the connections were fully functional (were supplying adequate quantity and providing a regular supply of water of potable quality) and approximately 80% of households received regular water supply as per the designated schedule.⁷ The average duration of water supplied to households was 3 hours per day.⁸

Figure 3: Improvement in the share of rural households with tap water connections under JJM by state and UT (2019-2023)



Source: Ministry of Jal Shakti, Jal Jeevan Mission Dashboard, accessed June 2023

³ Jal Jeevan Mission, Functional Household Tap Connection (FHTC) in every rural home, accessed 17th June 2023

⁴ Ministry of Jal Shakti, Tap Water Supply Reaches 66% Schools & 60% Anganwadi Centres, 2021

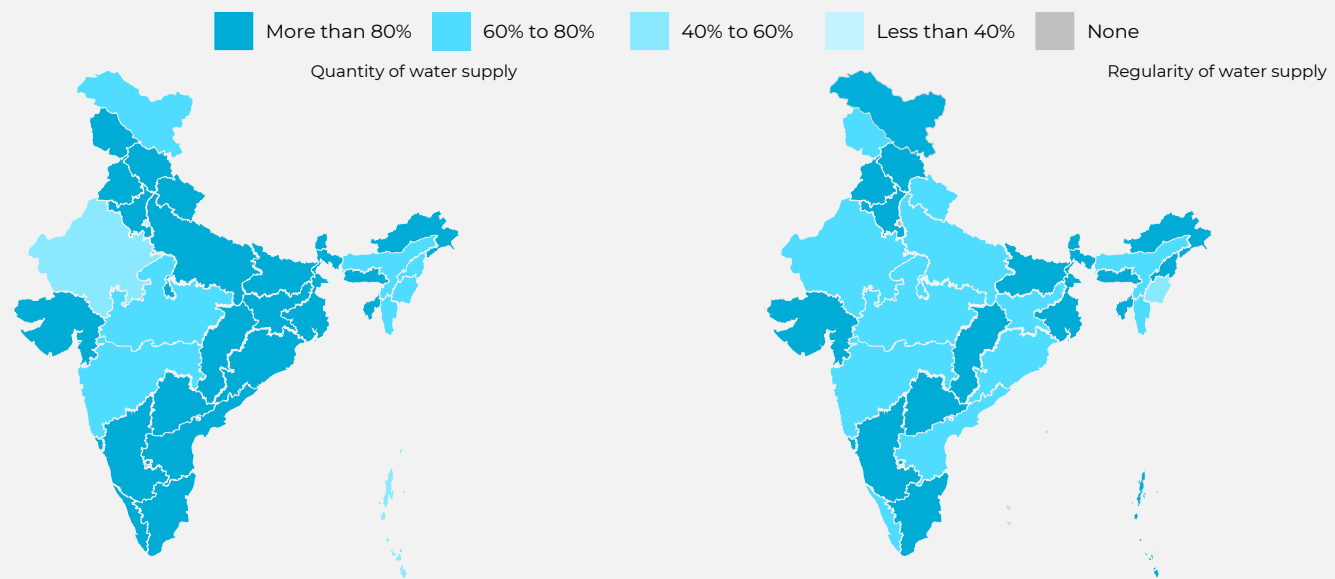
⁵ Ministry of Jal Shakti, Tap Water Supply Reaches 66% Schools & 60% Anganwadi Centres, 2021

⁶ Jal Jeevan Mission, Tap connection in schools/ AWCs/ GPs/ CHCs, accessed 17th June 2023

⁷ Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022; Fully Functional has been computed as = Adequate Quantity $\cap$ Fully Regular Supply $\cap$ Potable (Quality)

⁸ Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

Figure 4: Quantity and regularity of water received at the rural household level by state and UT (2022)



Source: Ministry of Jal Shakti, Functionality Assessment of Household Tap Connections, 2022

Water Quality

Water quality policies include a focus on improving infrastructure for water quality testing (testing labs and kits) and enhancing capacity building of specific individuals (e.g., Accredited Social Health Activists, VWSC members, and teachers) to undertake water testing and water treatment interventions.⁹

Rural water supply sources are contaminated:

85% of India's rural water supply is dependent on groundwater, while the remaining 15% comes from surface water.¹⁰ However, according to an assessment by the Central Ground Water Board in 2018, 52% of blocks in the country had at least one geogenic contaminant (arsenic, chloride, fluoride, iron, nitrate, and salinity).¹¹ Moreover, surface water sources in India are prone to pollution due to factors such as excessive use of chemical fertilizers, pesticides and discharge of industrial wastewater.¹² Mixing of human waste with drinking water sources poses health risks, and upstream towns and cities can contribute to this issue.

However, the potability and satisfaction with the quality of water supplied via tap water connections is high:

A recent report by JJM from 2022 noted that out of 86% of surveyed households with tap-water connections, 87% of households received potable water.¹³ 82% of households nationwide with tap-water connections reported being satisfied with the quality, 84% with the colour, and 83% with the taste of water received.¹⁴

Water testing capacities in the villages have shown improvement:

According to the JJM surveys, the number of villages with access to H2S vials increased sixfold from 2021-22 to 2022-23.¹⁵ Moreover, ~23 lakh women have been trained for testing water samples using field testing kits.¹⁶

9 Ministry of Jal Shakti, PIB, Quality of Tap Water Under Har Ghar Jal Yojana., 2022

10 Ministry of Jal Shakti, Drinking Water Quality Monitoring & Surveillance Framework, 2021; CGWB, Dynamic Ground Water Resources of India, 2022

11 Ministry of Jal Shakti, Drinking Water Quality Monitoring & Surveillance Framework, 2021

12 Ministry of Jal Shakti, Drinking Water Quality Monitoring & Surveillance Framework, 2021

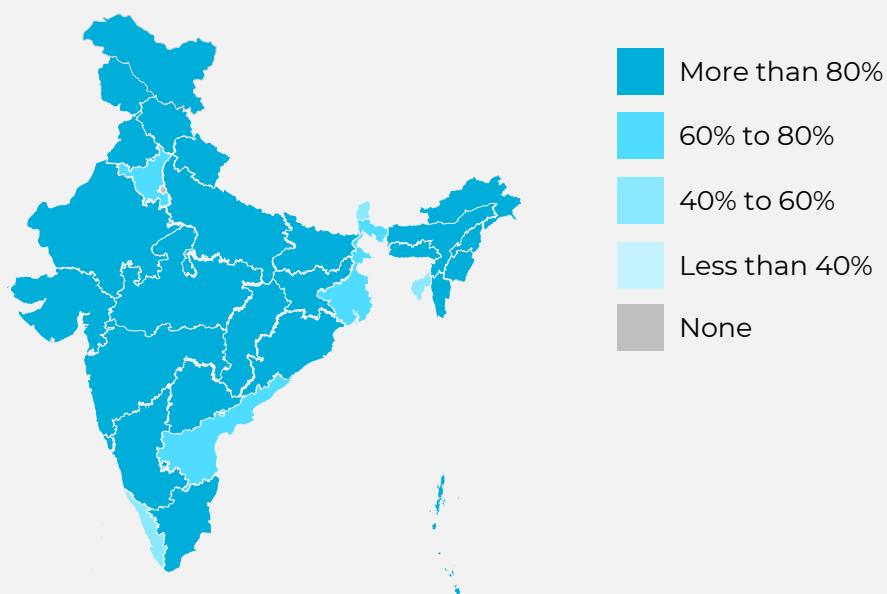
13 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

14 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022,

15 Jal Jeevan Mission, Water Quality Management Information System, 2023 (Note: H2S Strip test is a water quality testing kit used to detect bacteriological contamination)

16 Jal Jeevan Mission, Water Quality Management Information System, accessed August 2023

Figure 5: Potability of water received at the rural household level by state and UT (2022)



Source: Ministry of Jal Shakti, Functionality Assessment of Household Tap Connections, 2022

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by establishing quality rural tap water connections. These revolve primarily around improved health outcomes, better gender equity, and greater economic productivity.



Improved health outcomes and cost savings:

As per a WHO report highlighting the benefits of the 'Har Ghar Jal' program, it is estimated that ensuring safely managed drinking water for all Indian households could avert nearly 400,000 deaths due to diarrheal diseases and prevent ~14 million Disability Adjusted Life Years related to such diseases, which has the potential for supporting total cost saving up to USD 101 billion.¹⁷



Improved gender equity:

The responsibility of undertaking water-related tasks, including fetching, collecting, and purifying water, has traditionally fallen upon women in India. Improving drinking water access in the household through taps can help reduce such responsibilities and associated care burdens on women. The JJM Functionality Assessment of Household Tap Water Connections report highlights that out of households where female members used to fetch water before the installation of connections, 79% reported a reduction in drudgery in collecting water after the installation of such connections.¹⁸



Improved economic productivity:

As per the JJM Functionality Assessment of Household Tap Water Connections report, since the installation of a functional household tap connection, 31% of households reported that there had been a positive change in the number of employment days of the adult household members, and 66% of the households reported their income had directly benefitted.¹⁹ Expanding tap water connections to all rural households, hence, can help improve economic productivity.

Challenges

This section highlights some of the major challenges and constraints facing rural tap water supply in India. Efforts to resolve such challenges will ensure the timely establishment of tap connections, improved maintenance of water supply, and better water quality.



Procedural delays and limited capacity for O&M of piped water supply (PWS):

As per the Department of Drinking Water and Sanitation, states face several challenges in establishing tap connections which delay execution, such as uneven geographic terrains, scattered rural habitats, groundwater depletion, and adverse climatic conditions.²⁰ These are coupled with procedural challenges such as delays in obtaining clearances for project execution and the inability of states to provide a matching share in funds.²¹ Under JJM, GPs or their sub-committees such as the Village Water And Sanitation Committee (VWSC) and Pani Samiti are to play a key role in the planning, implementation, management, and O&M of in-village water

¹⁷ Ministry of Health & Family Welfare, PIB, WHO report highlights significant impact of 'Har Ghar Jal' Program on Public Health and Economic Savings, 2023

¹⁸ Department of Drinking Water and Sanitation, Functionality Assessment of Household Tap Connections, 2022

¹⁹ Department of Drinking Water and Sanitation, Functionality Assessment of Household Tap Connections, 2022

²⁰ Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

²¹ Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

supply.²² However, as per the Functionality Assessment of Household Tap Connections 2022, only 38% of sampled villages in India had constituted a VWSH or Pani Samiti.²³ Moreover, in only 14% of villages, the VWSC or Pani Samitis are responsible for the O&M of the PWS.²⁴ Overall, only ~31% of villages reported having identified skilled manpower for O&M of PWS.²⁵



Poor water quality and inadequate treatment regimes:

Chlorine treatment of water before it enters the piped water supply is one of the most important preventive actions to ensure water quality against bacteriological contamination.²⁶ According to a JJM Report, 72% of the collected water samples had no trace of residual chlorine, but in 4% of the cases, residual chlorine was beyond the permissible range.²⁷ Further, under JJM, states and union territories have been advised to set up Community Water Purification Plants (CWPP) in habitations where water is adversely impacted by contaminants such as arsenic and fluoride.²⁸ The Standing Committee on Water Resources 2022-23 noted that only ~6% of quality-affected habitations had been covered by such plants.²⁹

Institutional framework and policy landscape —

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee rural tap water supply.



Ministry of Jal Shakti³⁰

The Ministry of Jal Shakti is responsible for the implementation of the Jal Jeevan Mission. It also houses the Department of Drinking Water and Sanitation (DDWS), which oversees the provision of safe and adequate drinking water to households in rural India.



Ministry of Panchayati Raj

The Ministry is responsible for all matters related to the Panchayati Raj and Panchayati Raj Institutions in rural areas. It collaborates with the Ministry of Jal Shakti to implement initiatives for tap water supply.

Major policies and programs

This section highlights existing policies related to rural tap water supply and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI for States platform.

22 Department of Drinking Water & Sanitation, National Jal Jeevan Mission, 2021

23 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

24 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

25 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

26 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

27 Department of Drinking Water & Sanitation, Jal Jeevan Mission: Functionality Assessment of Household Tap Connections, 2022

28 Lok Sabha Secretariat, Standing Committee on Water Resources, 2023; Ministry of Jal Shakti, PIB, Jal Jeevan Mission: Addressing the problem of water contamination, 2021

29 Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

30 Ministry of Jal Shakti, About Us, Accessed December 2023



Jal Jeevan Mission³¹

The Jal Jeevan Mission, under its Har Ghar Jal Programme, aims to provide every rural household with affordable and regular access to an adequate supply of safe drinking water through taps. The mission has introduced advanced technologies to ensure continuous water supply, maintain water quality, and facilitate effective planning and monitoring. It has also decentralized the water supply programs by devolving powers to the Gram Panchayat and/ or its sub-committee to plan, implement, manage, operate, and maintain. The mission has also implemented the

National Water Policy³²

It aims to manage India's scarce water resources and ensure sustainable use of water. It consists of provisions that regulate the domestic supply of water in urban areas by outlining the sources of domestic water supply in urban areas, laying out rules for water treatment systems and promoting data collection initiatives.

³¹ Ministry of Jal Shakti, Operational guidelines for the implementation of JJM, 2019

³² Ministry of Jal Shakti, National Water Policy, 2012

Way forward

India has made significant progress in providing rural citizens with household tap connections. The coverage has increased significantly, and a large fraction of the households are satisfied with the quantity, regularity, and potability of water.

However, there are still key challenges: the averages hide how certain states have significantly lower coverage, how water quality is a key challenge in several locations with contaminants such as iron and fluoride, the water quality lab network needs strengthening, and the village water committees need to be expanded and made more effective.

Some ideas that could accelerate progress include:

 Developing helplines for rapid redressal and responses to water needs	Investments in regional helplines and helpdesks that can address water interruption or water quality complaints and also help gather highly granular data in terms of service delivery. AI tools, such as LLMs and Indian language translation can help make these helplines more accessible. Internet-of-Things solutions could also enable data capture at the community level. Partnerships with several social innovation firms with expertise in this could be explored.
 Establishing partnerships to expand coverage	Special attention should be paid to improving coverage in states that are lagging behind the average in terms of providing tap coverage. These states could consider partnerships with expert institutions to help provide networks in challenging areas in terms of topography or population densities. States should consider adopting from a broad menu of options, such as community-driven water supply networks, instead of having to always deploy from a standardized menu.
 Ensuring effective water continuity planning	Water source protection, both in terms of quantity and quality, will be critical going forward to ensure uninterrupted water supply in rural tap networks. This will become especially important in the face of accelerating climate change. States should prepare water continuity plans in the face of risks of floods and droughts. Cross-department collaborations that enable joint planning with watershed programs should be encouraged.

Recommended readings

Ministry of Jal Shakti. (2023). *Standing Committee on Water Resources – Demand for Grants (2022-23)*.
https://loksabhadocs.nic.in/lsscommittee/Water%20Resources/17_Water_Resources_19.pdf

World Health Organization. (2023). *Estimating potential health gains from increased access to safely managed drinking water services following the Jal Jeevan Mission initiative*.
<https://jaljeevanmission.gov.in/sites/default/files/2023-06/Jal-Jeevan-Mission-Summary-of-report.pdf>

Ministry of Jal Shakti. (2022). *Functionality Assessment of Household Tap Connections – Jal Jeevan Mission*.
https://jaljeevanmission.gov.in/sites/default/files/2022-10/national_report_of_functionality_assessment_2022.pdf

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